

Marco Ching

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Education

University of Waterloo - Waterloo, ON, Canada

09/2022 - Present

Honours Mechanical Engineering - Bachelor of Applied Science (BASc)

Skills

Mechanical Design & BIM: SolidWorks, AutoCAD, Revit, BIM, GD&T, DFM/A, Tolerance Stack-up, SketchUp

Building Systems & Energy Analysis: HVAC Design & Load Calculations, Fire Protection Design, U-Value & Envelope Heat Transfer Analysis, Lighting & Illuminance Calculations

Construction & Digital Engineering: Dynamo, Bluebeam, Autodesk Construction Cloud (ACC), ArcGIS, Power BI, Technical Schematics

Electronics & Controls: Raspberry Pi, Arduino, PLC, Sensor Integration, Motor Control, Multimeter, Oscilloscope

Prototyping & Manufacturing: 3D Printing, Hydraulic & Pneumatic Circuit Design, Laser Cutting, Fabrication (Mill, Lathe, Drill Press)

Programming: C++, MATLAB, Simulink

Work Experience

WSP Global Inc. - BIM Co-op Student - Toronto, Canada

01/2026 - 04/2026

- Developed and coordinated Revit BIM models for HVAC and plumbing systems on the Eglinton Crosstown West Extension preliminary design, producing ductwork, piping, air-terminal, valve, and fitting layouts across multiple views and sheets.
- Designed fire-protection sprinkler layouts and water-distribution systems for Union Station Expansion Project Blueroute Phases 2-6, supporting code-compliant building-services design and multidisciplinary coordination.
- Managed BIM coordination and submission schedules for OTA Development, tracking Internal, Interdisciplinary, Alliance, and IFC review milestones across 10 project scopes.

Airport Authority Hong Kong - BIM Engineering Trainee - Hong Kong

06/2025 - 08/2025

- Automated BIM and construction-data workflows using Dynamo, Power BI, and Autodesk Construction Cloud, supporting the integration of ACC into airport engineering and project-management processes.
- Reviewed and maintained 20+ Building Information Models in Revit for Hong Kong Airport Island, checking model accuracy and alignment with architectural and engineering standards.
- Applied SLAM and LiDAR technologies in an intern team project for real-time mapping and autonomous navigation of an AI-driven waste-sorting patrol robot.

Hong Kong Broadband Network - Engineering Intern - Hong Kong

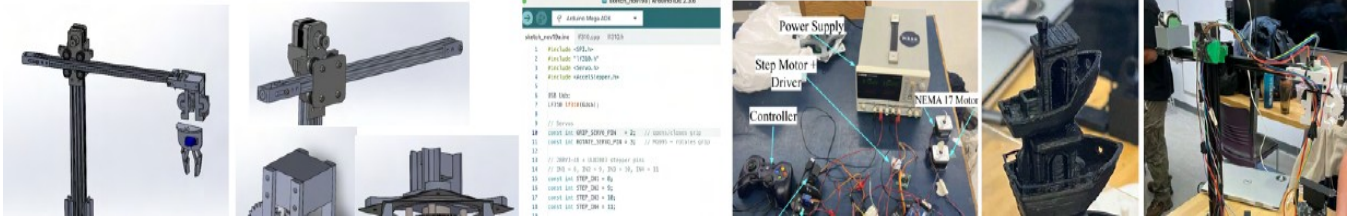
09/2024 - 12/2024

- Designed telecommunications infrastructure for 50+ residential projects by interpreting construction drawings and coordinating MDUs, fiber boxes, and 8-, 16-, 48-, and 72-core fiber-optic networks.
- Mapped and labeled 100+ network-administration components in ArcGIS Network Engineer across outdoor distribution systems, improving network documentation and asset organization.
- Supported senior management in allocating \$900,000 in funding across commercial and residential infrastructure projects, balancing technical requirements and production resources.

Portfolio

ULTRA PRECISE (+/-5 mm) Remote-Controlled ROBOTIC CRANE - 2025-2026

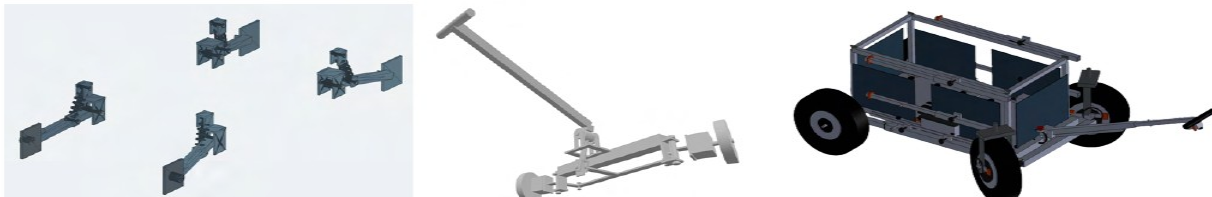
- Designed and fabricated a 5-DOF robotic manipulator in SolidWorks and 3D printed five custom mechanical components, achieving a 51 cm reach and 5 mm positional accuracy.
- Engineered motor-power and control circuitry for NEMA 17 steppers, a 28BYJ-48 stepper, and servo motors; diagnosed unreliable crimps, over-current, and overload issues during integration.
- Developed C++ firmware on an Arduino Mega ADK to map Logitech F310 controller inputs to coordinated joint motion and debugged USB host-shield communication faults.



CAPSTONE PROJECT: All-Terrain, Electronically-Assisted, Collapsible Wagon - Ongoing

Steering & Suspension System CAD in SolidWorks

- Designed a fail-safe mechanical steering system that converts lateral handle motion into front-wheel steering, maintaining directional control during an electrical or motor failure.
- Developed a standardized trailing-arm suspension layout for all four wheels using low-cost components to simplify fabrication and maintenance.
- Created and refined SolidWorks-compatible STEP/STL assemblies, checking fit, clearances, manufacturability, and system integration while progressing toward physical prototyping.



AWARD-WINNING Research on Volcanic Tectonic Earthquakes - 2024

IEEE Young Engineers Conference - 3rd Place Award

- Engineered and optimized a custom seismometer by integrating a geophone sensor with a Raspberry Pi, expanding the detection range from approximately 80 km to 100 km.
- Developed JavaScript-based data-visualization tools to generate seismographs from four months of seismic data.
- Presented the research at an IEEE conference, secured a \$500 sponsorship for continued investigation, and authored an investigative report published on the IEEE website.

